

Discussion 7

Consider two continuous random variables X and Y with joint p.d.f.

$$f_{X,Y}(x,y) = \begin{cases} C(x+2y) & 0 < x < 2, 0 < y < 3 \\ 0 & \text{otherwise} \end{cases}$$

- (1-1) Find the value of C so that $f_{X,Y}(x,y)$ is a valid joint p.d.f.
- (1-2) Find the marginal probability density function of X , $f_X(x)$.
- (1-3) Find the marginal probability density function of Y , $f_Y(y)$.
- (1-4) Are X and Y independent? If X and Y are not independent, find $Cov(X,Y)$.

2. A random variable X has a Exponential distribution with mean 0.2. Find $P(\mu - \sigma < X < \mu + \sigma)$.

3. 60 scientists from 30 universities attend a conference. The conference includes a lunch in a cafeteria which has 30 tables each suitable for 2 people. The people seated for lunch at random. Let $X_j = 1$ if the scientists from university j seat together and $X_j = 0$ otherwise.

- (a) Compute $E(X_1)$ and $V(X_1)$.
- (b) Compute $Cov(X_1, X_2)$.
- (c) Let $X = X_1 + X_2 + \cdots + X_{30}$ be the number of tables occupied by the people from the same university. Compute $E[X]$ and $V(X)$.

4. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} \frac{15}{2}x(2-x-y) & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Compute the conditional density of X , given that $Y = y$, where $0 < y < 1$.

5.

$x = 1$	$\frac{1}{6}$	$\frac{1}{8}$	$\frac{1}{4}$
$x = 0$	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{4}$
	$y = 0$	$y = 1$	$y = 2$

- (a) Find $P(X = 0|Y \leq 1)$.
- (b) Find the marginal PMFs of X and Y .
- (c) Find $P(Y = 1|X = 0)$.
- (d) Are X and Y independent?